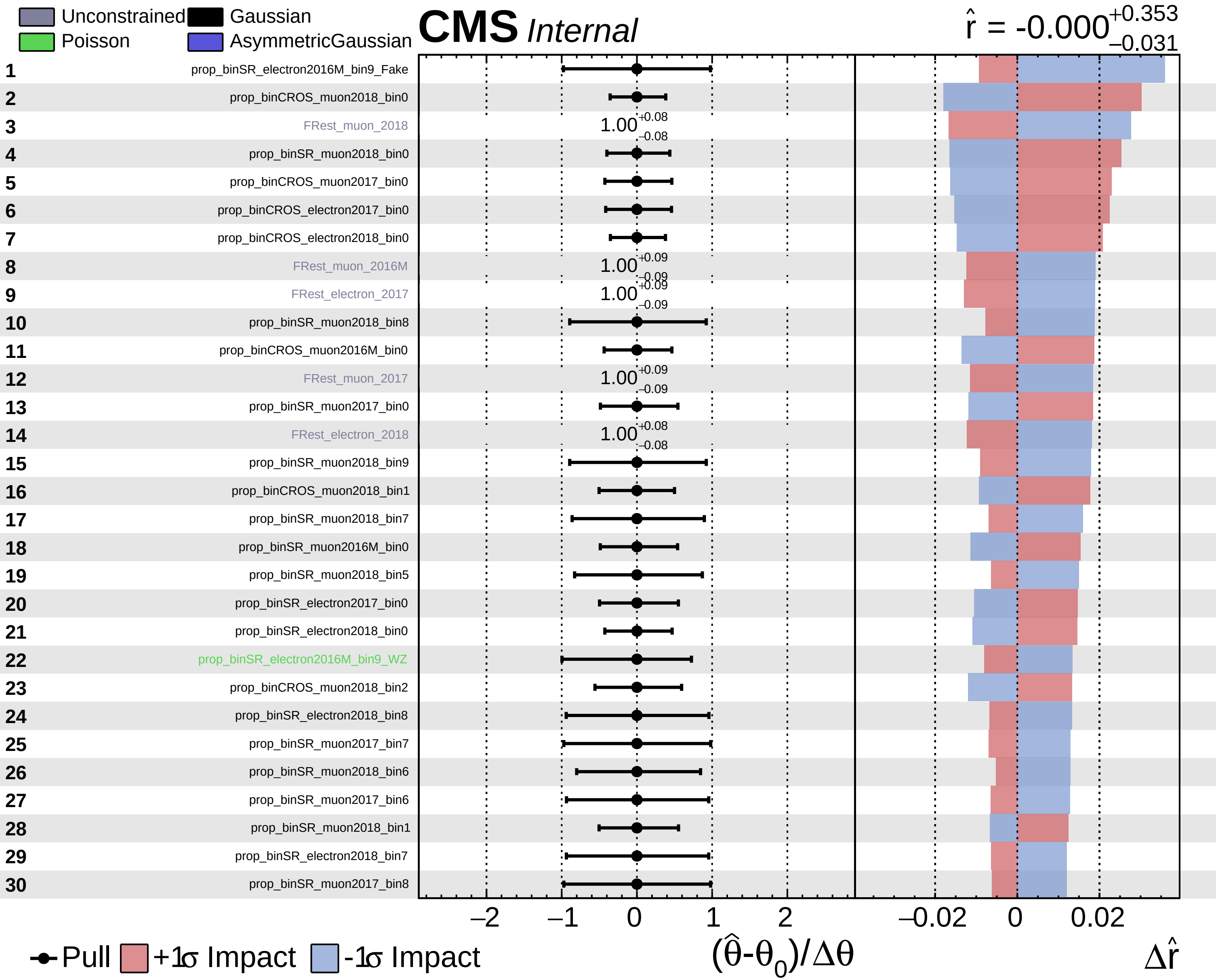
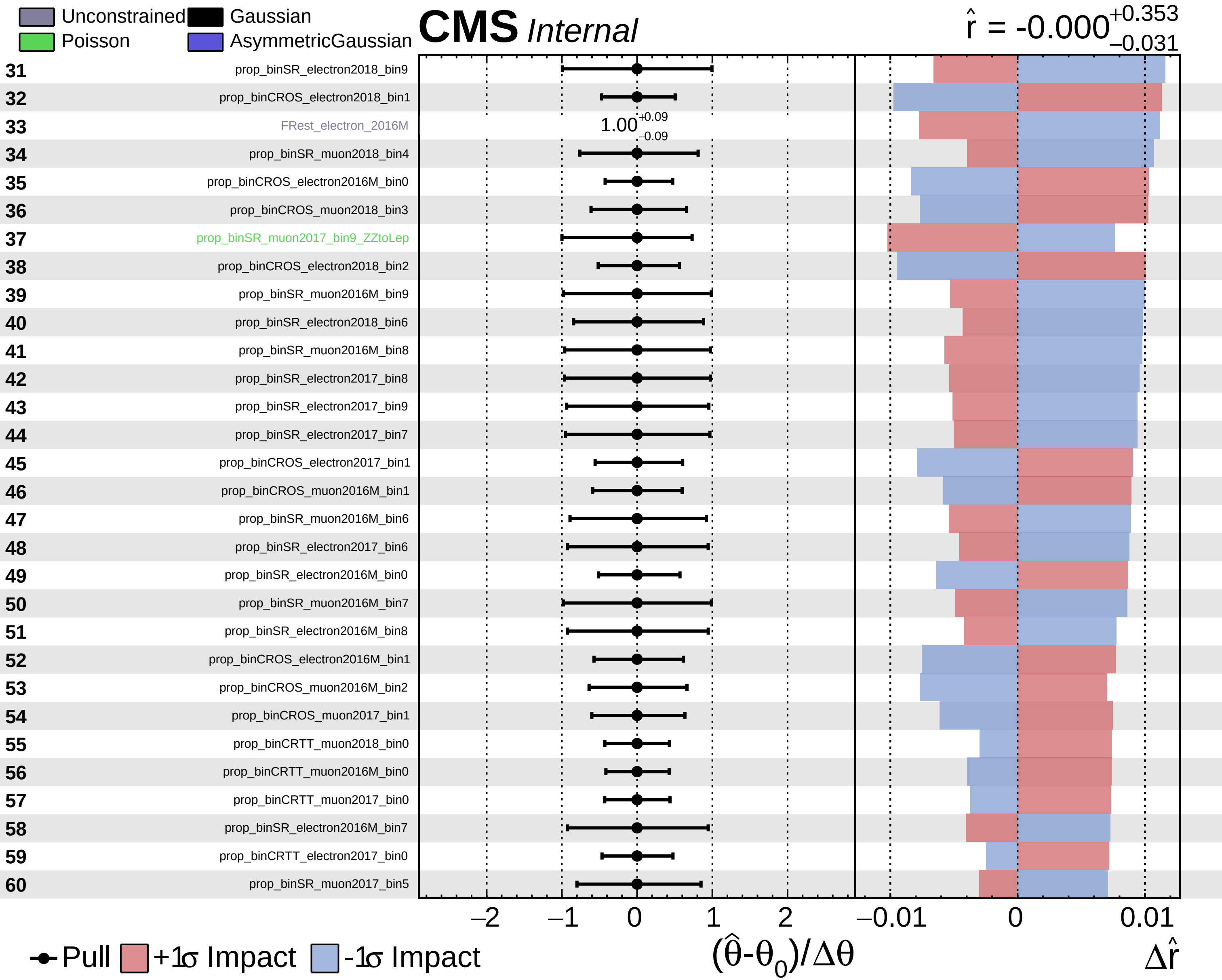
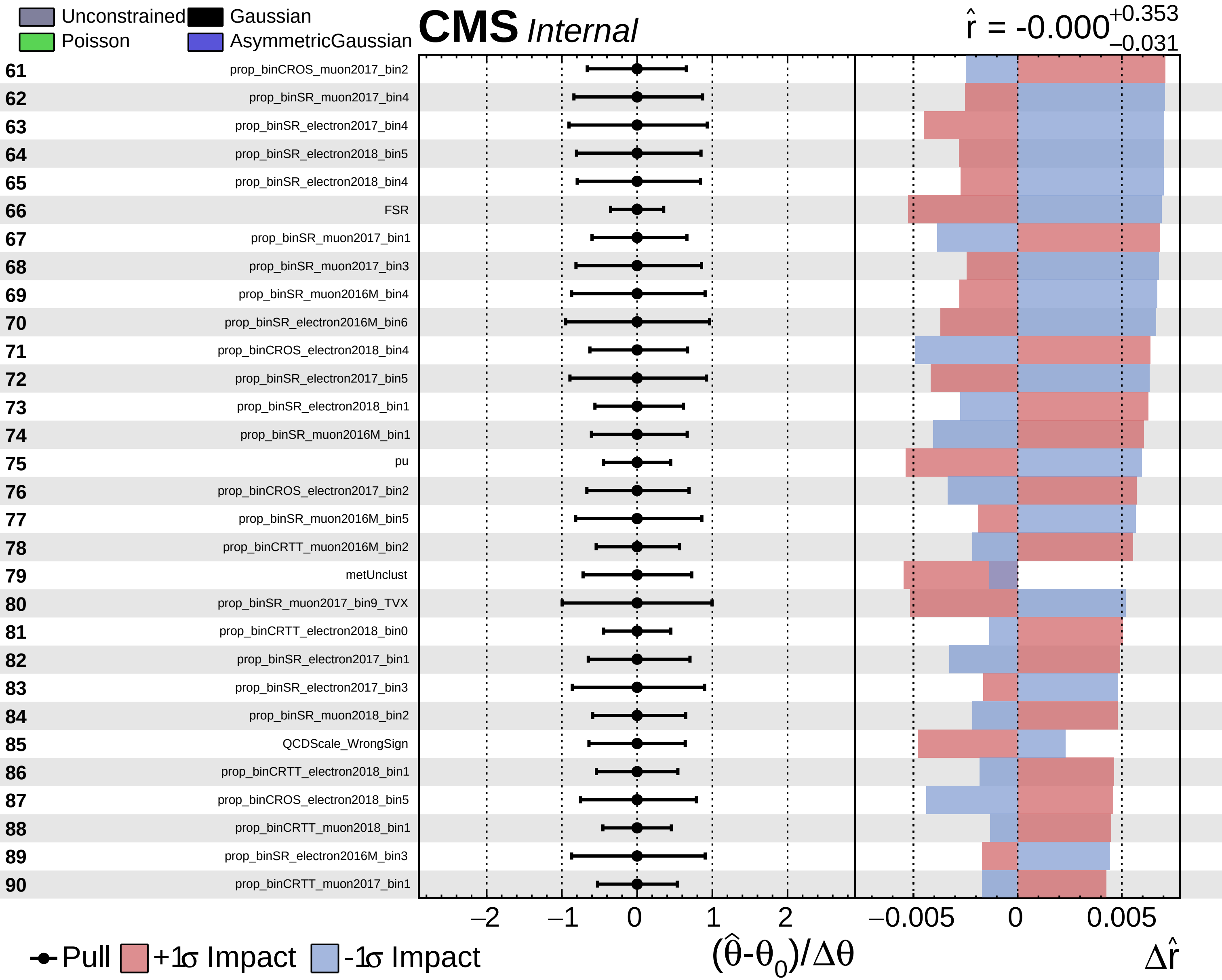


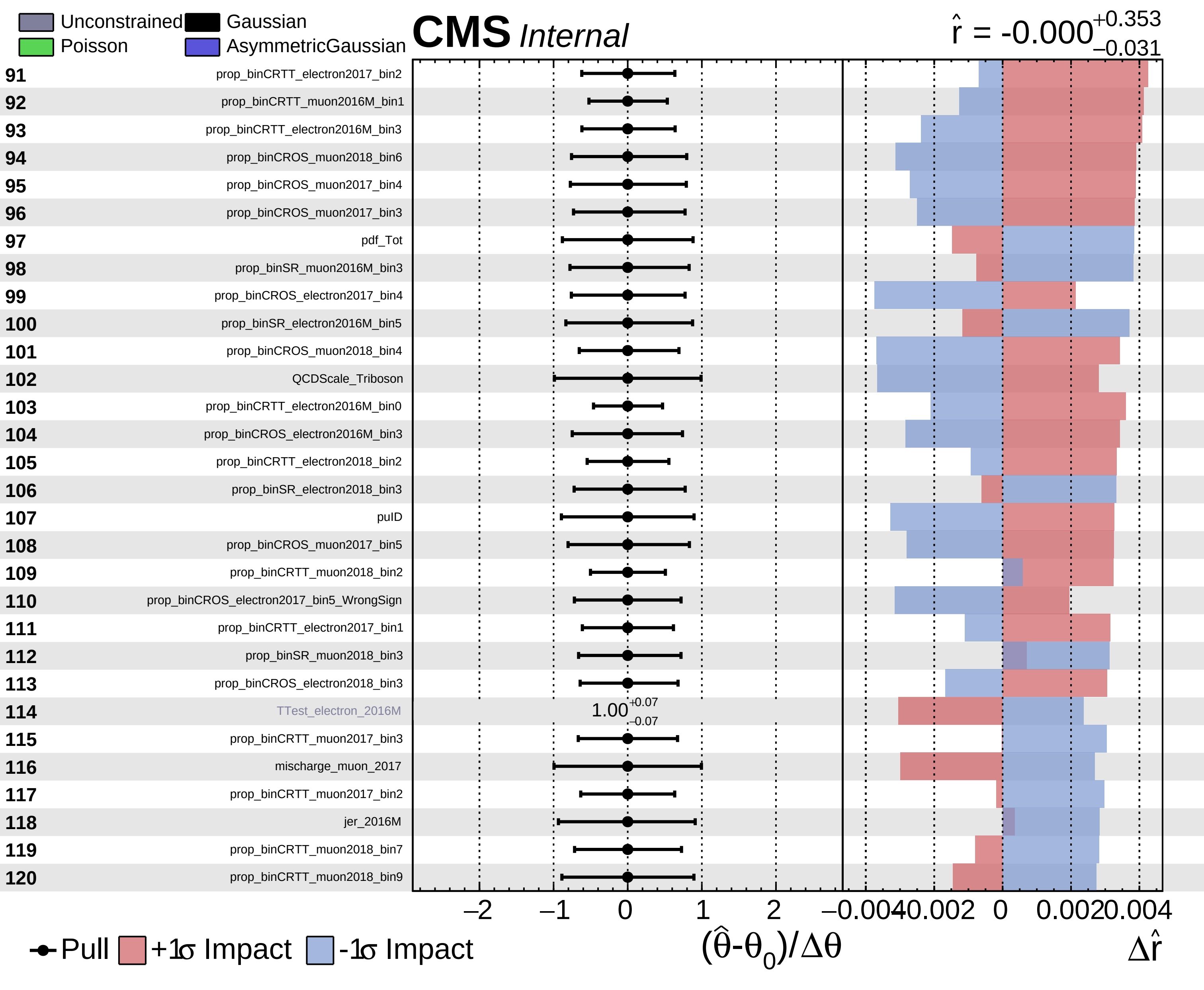
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.000 +0.353 −0.031
1 prop_binSR_electron2016M_bin9_Fake
2 prop_binCROS_muon2018_bin0
3 FRest_muon_2018 1.00 +0.08 −0.08
4 prop_binSR_muon2018_bin0
5 prop_binCROS_muon2017_bin0
6 prop_binCROS_electron2017_bin0
7 prop_binCROS_electron2018_bin0
8 FRest_muon_2016M 1.00 +0.09 −0.09
9 FRest_electron_2017 1.00 +0.09 −0.09
10 prop_binSR_muon2018_bin8
11 prop_binCROS_muon2016M_bin0
12 FRest_muon_2017 1.00 +0.09 −0.09
13 prop_binSR_muon2017_bin0
14 FRest_electron_2018 1.00 +0.08 −0.08
15 prop_binSR_muon2018_bin9
16 prop_binCROS_muon2018_bin1
17 prop_binSR_muon2018_bin7
18 prop_binSR_muon2016M_bin0
19 prop_binSR_muon2018_bin5
20 prop_binSR_electron2017_bin0
21 prop_binSR_electron2018_bin0
22 prop_binSR_electron2016M_bin9_WZ
23 prop_binCROS_muon2018_bin2
24 prop_binSR_electron2018_bin8
25 prop_binSR_muon2017_bin7
26 prop_binSR_muon2018_bin6
27 prop_binSR_muon2017_bin6
28 prop_binSR_muon2018_bin1
29 prop_binSR_electron2018_bin7
30 prop_binSR_muon2017_bin8
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



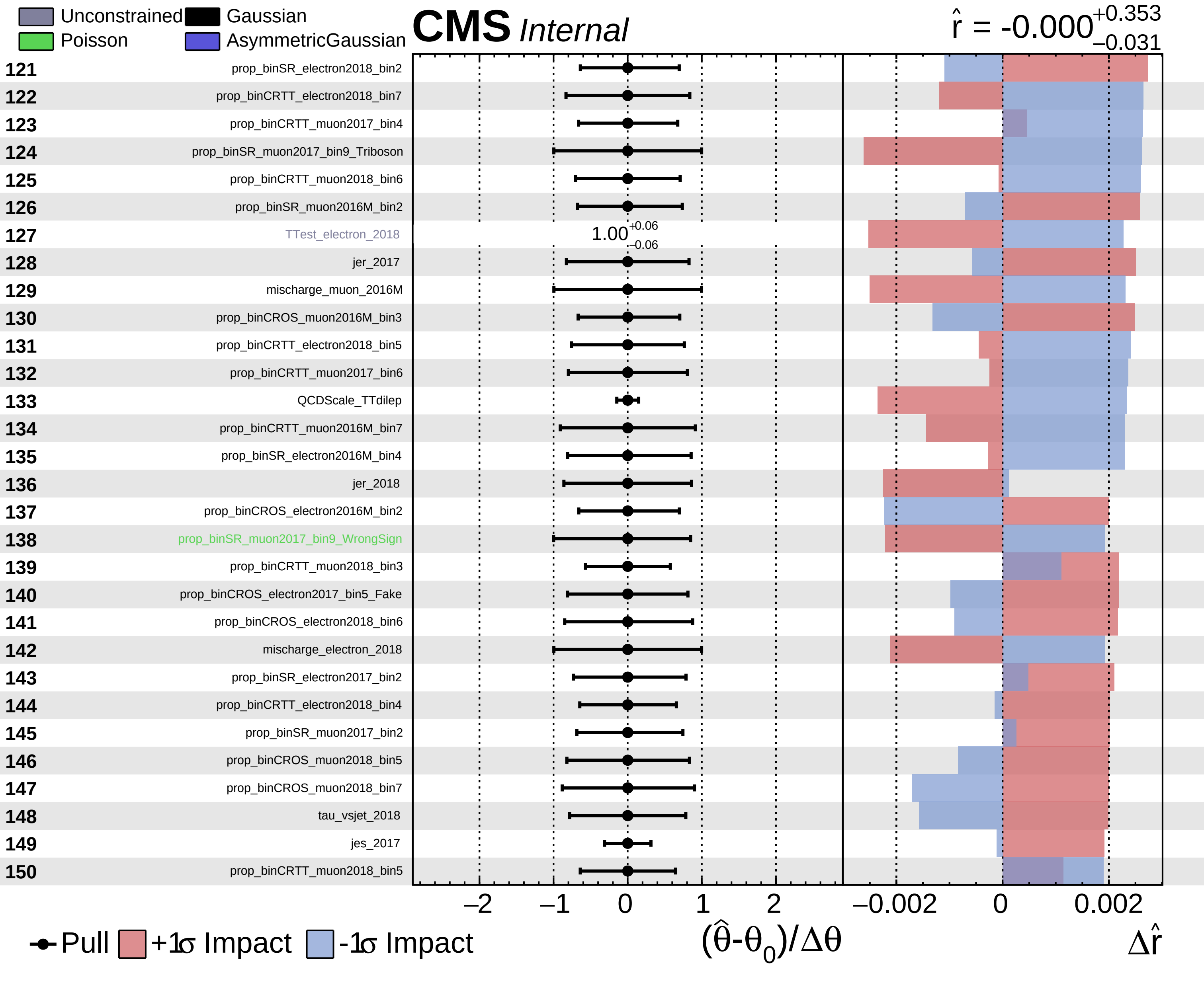
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.353}_{-0.031}$
31 prop_binSR_electron2018_bin9
32 prop_binCROS_electron2018_bin1
33 FRest_electron_2016M
$1.00^{+0.09}_{-0.09}$
34 prop_binSR_muon2018_bin4
35 prop_binCROS_electron2016M_bin0
36 prop_binCROS_muon2018_bin3
37 prop_binSR_muon2017_bin9_ZZtoLep
38 prop_binCROS_electron2018_bin2
39 prop_binSR_muon2016M_bin9
40 prop_binSR_electron2018_bin6
41 prop_binSR_muon2016M_bin8
42 prop_binSR_electron2017_bin8
43 prop_binSR_electron2017_bin9
44 prop_binSR_electron2017_bin7
45 prop_binCROS_electron2017_bin1
46 prop_binCROS_muon2016M_bin1
47 prop_binSR_muon2016M_bin6
48 prop_binSR_electron2017_bin6
49 prop_binSR_electron2016M_bin0
50 prop_binSR_muon2016M_bin7
51 prop_binSR_electron2016M_bin8
52 prop_binCROS_electron2016M_bin1
53 prop_binCROS_muon2016M_bin2
54 prop_binCROS_muon2017_bin1
55 prop_binCRTT_muon2018_bin0
56 prop_binCRTT_muon2016M_bin0
57 prop_binCRTT_muon2017_bin0
58 prop_binSR_electron2016M_bin7
59 prop_binCRTT_electron2017_bin0
60 prop_binSR_muon2017_bin5
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



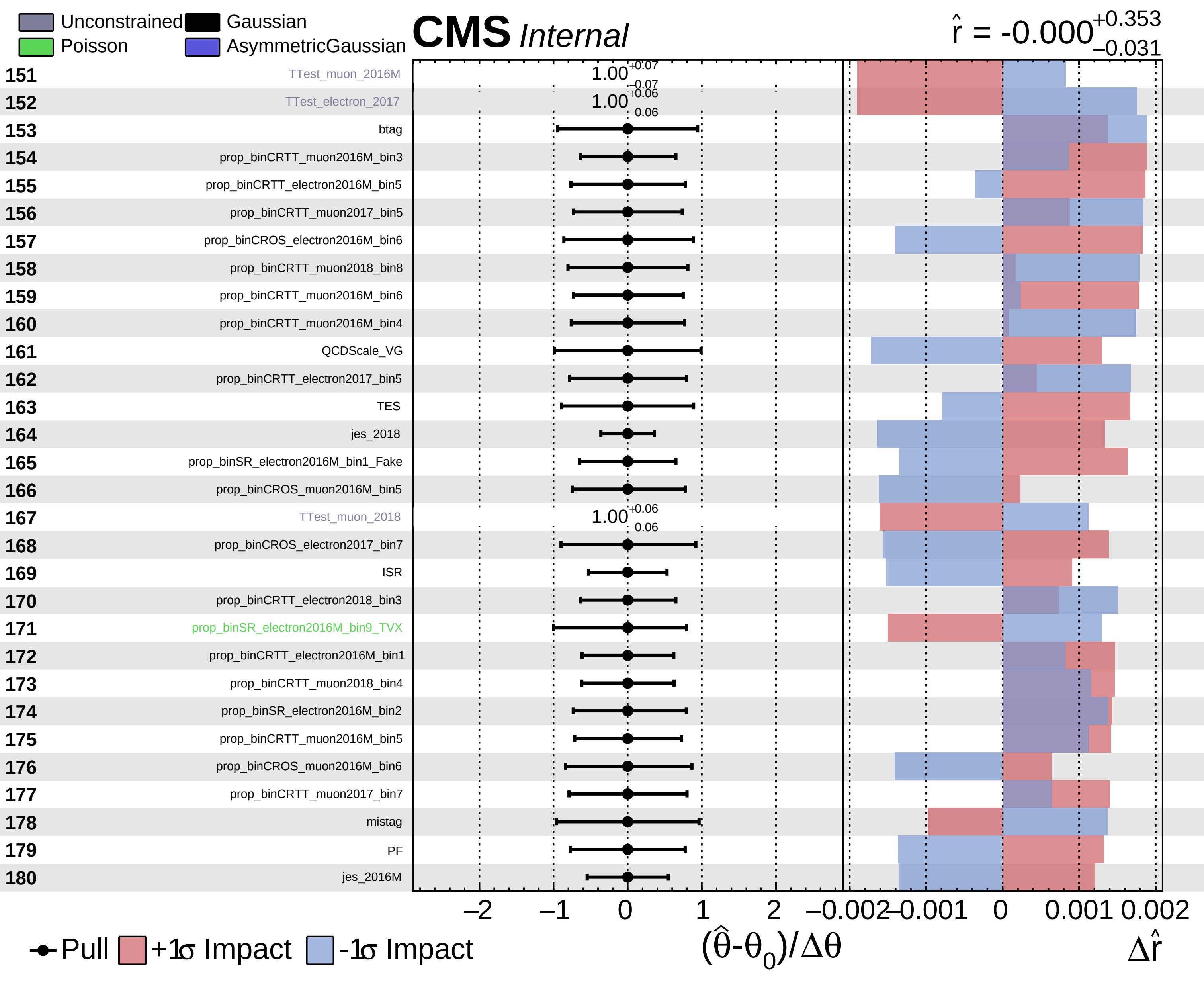
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.353}_{-0.031}$
61 prop_binCROS_muon2017_bin2
62 prop_binSR_muon2017_bin4
63 prop_binSR_electron2017_bin4
64 prop_binSR_electron2018_bin5
65 prop_binSR_electron2018_bin4
66 FSR
67 prop_binSR_muon2017_bin1
68 prop_binSR_muon2017_bin3
69 prop_binSR_muon2016M_bin4
70 prop_binSR_electron2016M_bin6
71 prop_binCROS_electron2018_bin4
72 prop_binSR_electron2017_bin5
73 prop_binSR_electron2018_bin1
74 prop_binSR_muon2016M_bin1
75 pu
76 prop_binCROS_electron2017_bin2
77 prop_binSR_muon2016M_bin5
78 prop_binCRTT_muon2016M_bin2
79 metUnclust
80 prop_binSR_muon2017_bin9_TVX
81 prop_binCRTT_electron2018_bin0
82 prop_binSR_electron2017_bin1
83 prop_binSR_electron2017_bin3
84 prop_binSR_muon2018_bin2
85 QCDScale_WrongSign
86 prop_binCRTT_electron2018_bin1
87 prop_binCROS_electron2018_bin5
88 prop_binCRTT_muon2018_bin1
89 prop_binSR_electron2016M_bin3
90 prop_binCRTT_muon2017_bin1
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



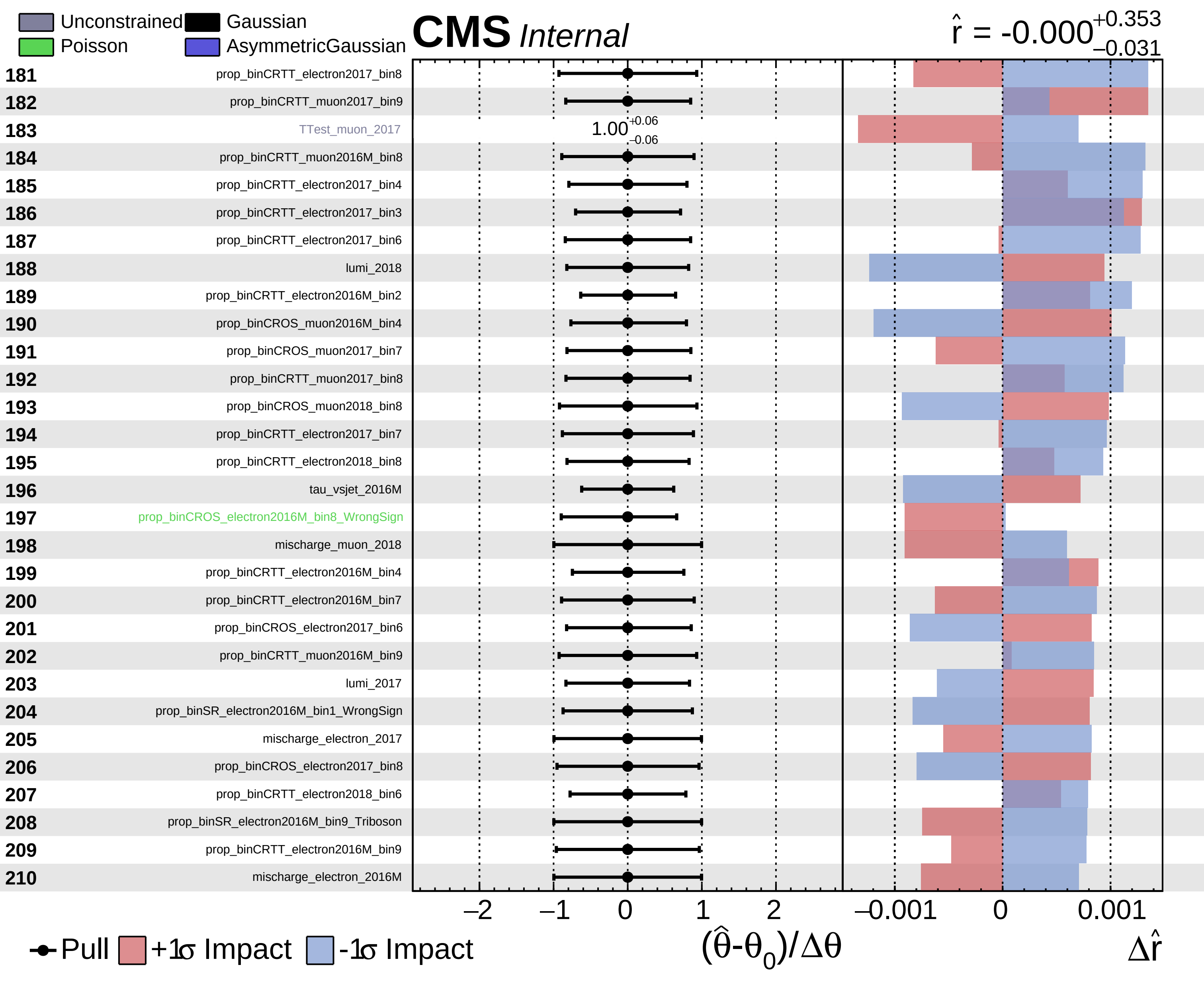
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.000 +0.353 −0.031
91 prop_binCRTT_electron2017_bin2
92 prop_binCRTT_muon2016M_bin1
93 prop_binCRTT_electron2016M_bin3
94 prop_binCROS_muon2018_bin6
95 prop_binCROS_muon2017_bin4
96 prop_binCROS_muon2017_bin3
97 pdf_Tot
98 prop_binSR_muon2016M_bin3
99 prop_binCROS_electron2017_bin4
100 prop_binSR_electron2016M_bin5
101 prop_binCROS_muon2018_bin4
102 QCDScale_Triboson
103 prop_binCRTT_electron2016M_bin0
104 prop_binCROS_electron2016M_bin3
105 prop_binCRTT_electron2018_bin2
106 prop_binSR_electron2018_bin3
107 puID
108 prop_binCROS_muon2017_bin5
109 prop_binCRTT_muon2018_bin2
110 prop_binCROS_electron2017_bin5_WrongSign
111 prop_binCRTT_electron2017_bin1
112 prop_binSR_muon2018_bin3
113 prop_binCROS_electron2018_bin3
114 TTest_electron_2016M
1.00 +0.07 −0.07
115 prop_binCRTT_muon2017_bin3
116 mischarge_muon_2017
117 prop_binCRTT_muon2017_bin2
118 jer_2016M
119 prop_binCRTT_muon2018_bin7
120 prop_binCRTT_muon2018_bin9
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



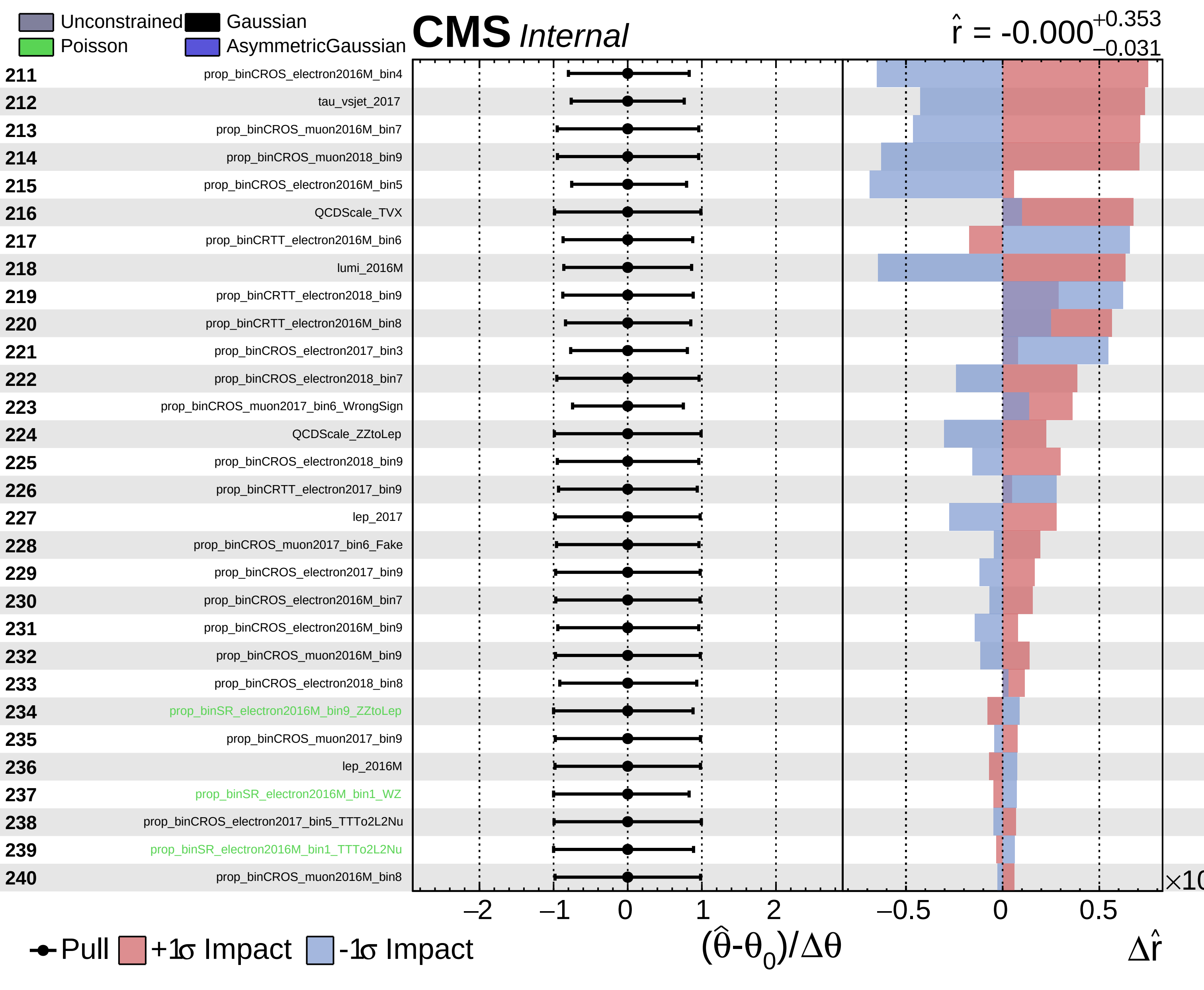
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.353}_{-0.031}$
121 prop_binSR_electron2018_bin2
122 prop_binCRTT_electron2018_bin7
123 prop_binCRTT_muon2017_bin4
124 prop_binSR_muon2017_bin9_Triboson
125 prop_binCRTT_muon2018_bin6
126 prop_binSR_muon2016M_bin2
127 TTest_electron_2018
$1.00^{+0.06}_{-0.06}$
128 jer_2017
129 mischarge_muon_2016M
130 prop_binCROS_muon2016M_bin3
131 prop_binCRTT_electron2018_bin5
132 prop_binCRTT_muon2017_bin6
133 QCDScale_TTdilep
134 prop_binCRTT_muon2016M_bin7
135 prop_binSR_electron2016M_bin4
136 jer_2018
137 prop_binCROS_electron2016M_bin2
138 prop_binSR_muon2017_bin9_WrongSign
139 prop_binCRTT_muon2018_bin3
140 prop_binCROS_electron2017_bin5_Fake
141 prop_binCROS_electron2018_bin6
142 mischarge_electron_2018
143 prop_binSR_electron2017_bin2
144 prop_binCRTT_electron2018_bin4
145 prop_binSR_muon2017_bin2
146 prop_binCROS_muon2018_bin5
147 prop_binCROS_muon2018_bin7
148 tau_vsjet_2018
149 jes_2017
150 prop_binCRTT_muon2018_bin5
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.353}_{-0.031}$
151 TTest_muon_2016M 1.00 +0.07 −0.07
152 TTest_electron_2017 1.00 +0.06 −0.06
153 btag
154 prop_binCRTT_muon2016M_bin3
155 prop_binCRTT_electron2016M_bin5
156 prop_binCRTT_muon2017_bin5
157 prop_binCROS_electron2016M_bin6
158 prop_binCRTT_muon2018_bin8
159 prop_binCRTT_muon2016M_bin6
160 prop_binCRTT_muon2016M_bin4
161 QCDScale_VG
162 prop_binCRTT_electron2017_bin5
163 TES
164 jes_2018
165 prop_binSR_electron2016M_bin1_Fake
166 prop_binCROS_muon2016M_bin5
167 TTest_muon_2018 1.00 +0.06 −0.06
168 prop_binCROS_electron2017_bin7
169 ISR
170 prop_binCRTT_electron2018_bin3
171 prop_binSR_electron2016M_bin9_TVX
172 prop_binCRTT_electron2016M_bin1
173 prop_binCRTT_muon2018_bin4
174 prop_binSR_electron2016M_bin2
175 prop_binCRTT_muon2016M_bin5
176 prop_binCROS_muon2016M_bin6
177 prop_binCRTT_muon2017_bin7
178 mistag
179 PF
180 jes_2016M
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.353}_{-0.031}$
181 prop_binCRTT_electron2017_bin8
182 prop_binCRTT_muon2017_bin9
183 TTest_muon_2017
$1.00^{+0.06}_{-0.06}$
184 prop_binCRTT_muon2016M_bin8
185 prop_binCRTT_electron2017_bin4
186 prop_binCRTT_electron2017_bin3
187 prop_binCRTT_electron2017_bin6
188 lumi_2018
189 prop_binCRTT_electron2016M_bin2
190 prop_binCROS_muon2016M_bin4
191 prop_binCROS_muon2017_bin7
192 prop_binCRTT_muon2017_bin8
193 prop_binCROS_muon2018_bin8
194 prop_binCRTT_electron2017_bin7
195 prop_binCRTT_electron2018_bin8
196 tau_vsjet_2016M
197 prop_binCROS_electron2016M_bin8_WrongSign
198 mischarge_muon_2018
199 prop_binCRTT_electron2016M_bin4
200 prop_binCRTT_electron2016M_bin7
201 prop_binCROS_electron2017_bin6
202 prop_binCRTT_muon2016M_bin9
203 lumi_2017
204 prop_binSR_electron2016M_bin1_WrongSign
205 mischarge_electron_2017
206 prop_binCROS_electron2017_bin8
207 prop_binCRTT_electron2018_bin6
208 prop_binSR_electron2016M_bin9_Triboson
209 prop_binCRTT_electron2016M_bin9
210 mischarge_electron_2016M
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.353}_{-0.031}$
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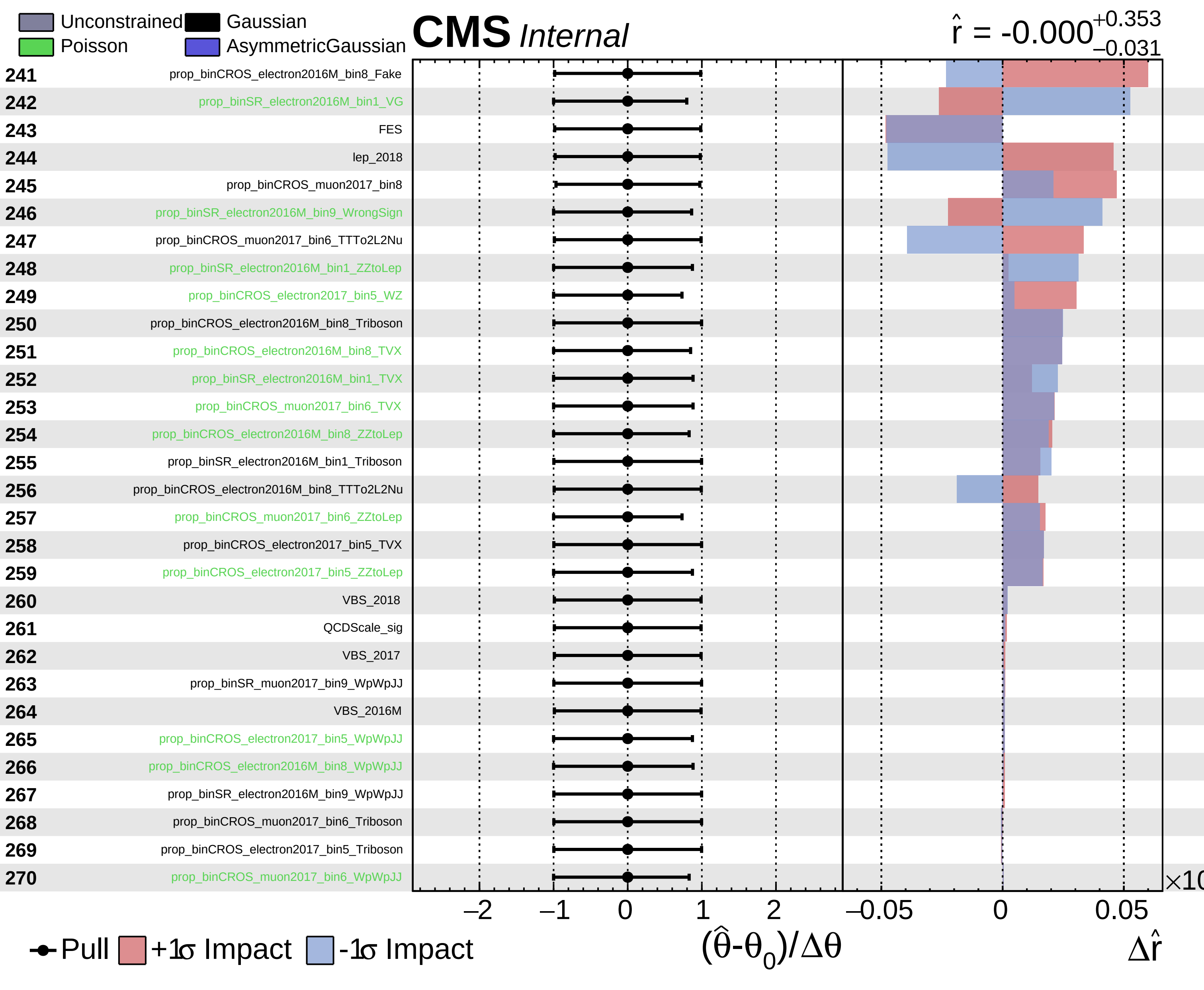
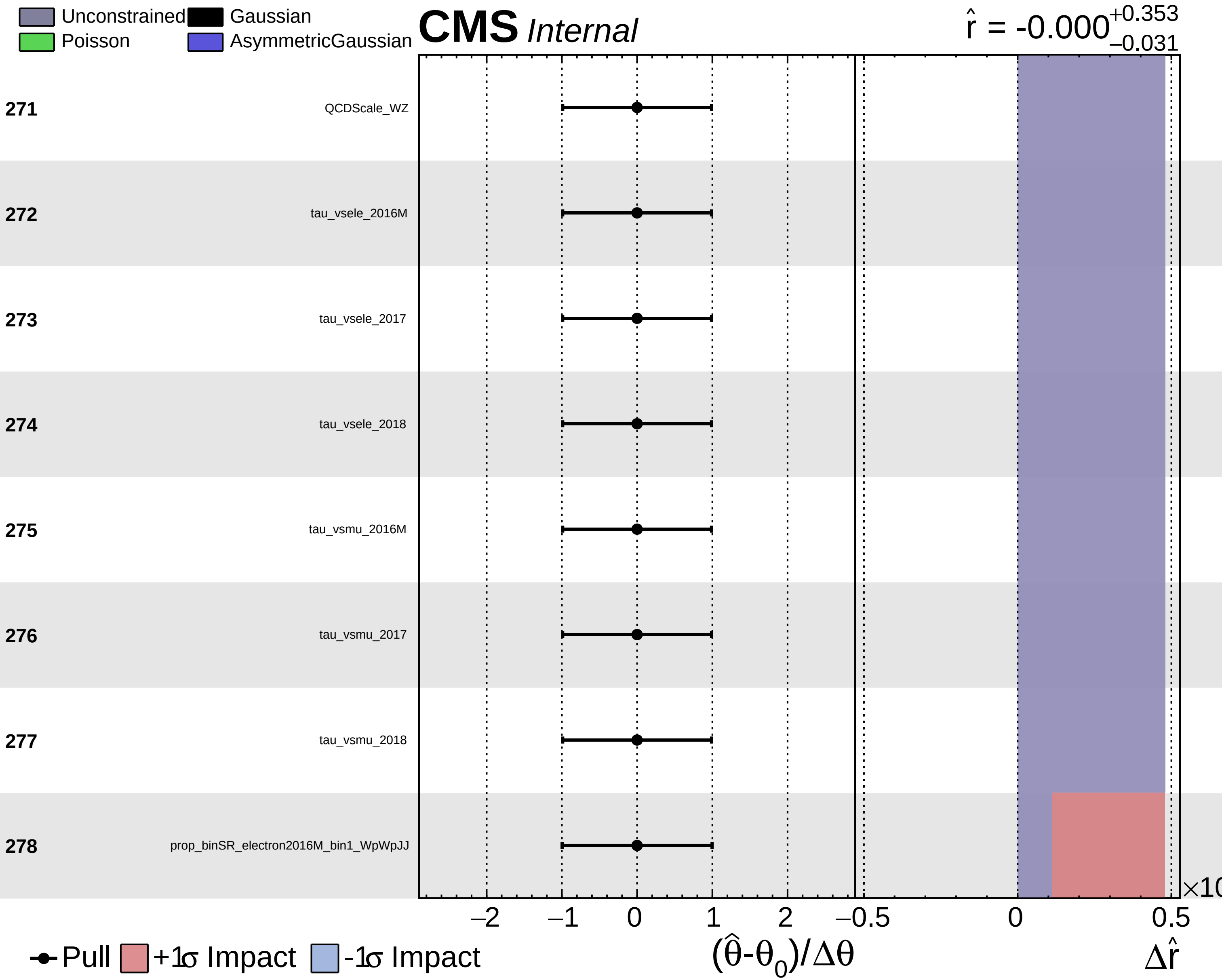
212 tau_vsjet_2017
213 prop_binCROS_muon2016M_bin7
214 prop_binCROS_muon2018_bin9
215 prop_binCROS_electron2016M_bin5
216 QCDScale_TVX
217 prop_binCRTT_electron2016M_bin6
218 lumi_2016M
219 prop_binCRTT_electron2018_bin9
220 prop_binCRTT_electron2016M_bin8
221 prop_binCROS_electron2017_bin3
222 prop_binCROS_electron2018_bin7
223 prop_binCROS_muon2017_bin6_WrongSign
224 QCDScale_ZZtoLep
225 prop_binCROS_electron2018_bin9
226 prop_binCRTT_electron2017_bin9
227 lep_2017
228 prop_binCROS_muon2017_bin6_Fake
229 prop_binCROS_electron2017_bin9
230 prop_binCROS_electron2016M_bin7
231 prop_binCROS_electron2016M_bin9
232 prop_binCROS_muon2016M_bin9
233 prop_binCROS_electron2018_bin8
234 prop_binSR_electron2016M_bin9_ZZtoLep
235 prop_binCROS_muon2017_bin9
236 lep_2016M
237 prop_binSR_electron2016M_bin1_WZ
238 prop_binCROS_electron2017_bin5_TTTo2L2Nu
239 prop_binSR_electron2016M_bin1_TTTo2L2Nu
240 prop_binCROS_muon2016M_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.353}_{-0.031}$
241 prop_binCROS_electron2016M_bin8_Fake
242 prop_binSR_electron2016M_bin1_VG
243 FES
244 lep_2018
245 prop_binCROS_muon2017_bin8
246 prop_binSR_electron2016M_bin9_WrongSign
247 prop_binCROS_muon2017_bin6_TTTo2L2Nu
248 prop_binSR_electron2016M_bin1_ZZtoLep
249 prop_binCROS_electron2017_bin5_WZ
250 prop_binCROS_electron2016M_bin8_Triboson
251 prop_binCROS_electron2016M_bin8_TVX
252 prop_binSR_electron2016M_bin1_TVX
253 prop_binCROS_muon2017_bin6_TVX
254 prop_binCROS_electron2016M_bin8_ZZtoLep
255 prop_binSR_electron2016M_bin1_Triboson
256 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
257 prop_binCROS_muon2017_bin6_ZZtoLep
258 prop_binCROS_electron2017_bin5_TVX
259 prop_binCROS_electron2017_bin5_ZZtoLep
260 VBS_2018
261 QCDScale_sig
262 VBS_2017
263 prop_binSR_muon2017_bin9_WpWpJJ
264 VBS_2016M
265 prop_binCROS_electron2017_bin5_WpWpJJ
266 prop_binCROS_electron2016M_bin8_WpWpJJ
267 prop_binSR_electron2016M_bin9_WpWpJJ
268 prop_binCROS_muon2017_bin6_Triboson
269 prop_binCROS_electron2017_bin5_Triboson
270 prop_binCROS_muon2017_bin6_WpWpJJ
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.353}_{-0.031}$
271 QCDScale_WZ
272 tau_vsele_2016M
273 tau_vsele_2017
274 tau_vsele_2018
275 tau_vsmu_2016M
276 tau_vsmu_2017
277 tau_vsmu_2018
278 prop_binSR_electron2016M_bin1_WpWpJJ
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$